

HashSet Interview Handbook

1. Why was HashSet created?

Suppose you need to store millions of usernames and never allow duplicates. Searching an ArrayList before every insertion is $O(n)$. HashSet was created to solve this by providing fast duplicate detection using hashing.

2. Main Goal

HashSet has one responsibility: store only unique elements. It does not store values against keys and does not provide index-based access.

3. Internal Working

HashSet internally uses a HashMap.

Your element becomes the **key**.

Java stores a single dummy object called **PRESENT** as the value.

```
HashSet.add("Rahul")
```

↓

```
HashMap.put("Rahul", PRESENT)
```

4. How add() Works

Step 1: Call `hashCode()` on the element.

Step 2: Convert hash into a bucket (array index).

Step 3: Go to that bucket.

Step 4: If collision occurs, compare keys using `equals()`.

Step 5: If an equal key already exists, insertion is ignored. Otherwise a new node is added.

5. How Duplicates Are Prevented

`hashCode()` finds the bucket quickly. `equals()` confirms whether the existing element and the new element are logically the same. Both methods work together.

6. Custom Objects

If `Student(id=1,name='Rahul')` and another `Student(id=1,name='Rahul')` should be treated as the same student, override BOTH `equals()` and `hashCode()`. Overriding only one of them can produce incorrect behavior.

7. Time Complexity

Operation	Average	Reason
<code>add()</code>	$O(1)$	Hashing
<code>contains()</code>	$O(1)$	Direct bucket lookup
<code>remove()</code>	$O(1)$	Same hashing mechanism

8. HashSet vs ArrayList

ArrayList: allows duplicates, maintains order, supports index.

HashSet: unique elements only, no index, no guaranteed iteration order, faster duplicate checks.

9. Common Interview Questions

- Why doesn't HashSet allow duplicates?
- How does HashSet use HashMap internally?
- Why override equals() and hashCode()?

10. Common Mistakes

- Thinking HashSet has its own storage implementation.
- Forgetting it internally uses HashMap.
- Overriding equals() without hashCode().

11. Remember Forever

HashSet = HashMap wrapper.

Element = HashMap Key.

PRESENT = Dummy Value.

hashCode() → Bucket.

equals() → Duplicate check.